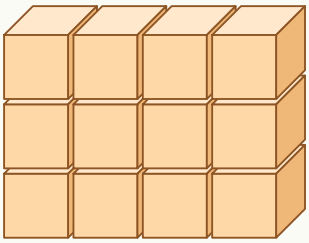
Chapter 5

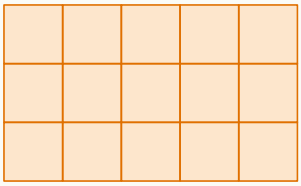
Measurement

how long, how much space, how much stuff, how fast





12 little cubes: a volume



15 squares: an area

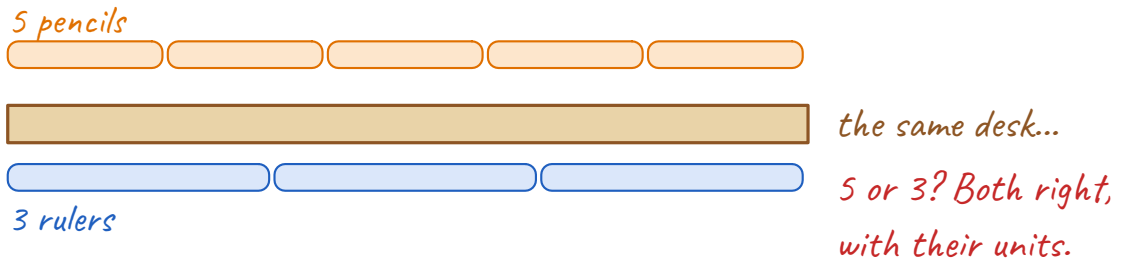
Measuring is counting how many **units** fit. A length counts little rulers, an area counts little squares, a volume counts little cubes. Almost every mistake in this topic comes from mixing up units, so this chapter slows right down on **changing units**.

It also covers perimeter, area and volume of the everyday shapes, time (which is sneakily not decimal), and "per" units like metres per second, which appear all over your lectures (the slides write speeds as m s^{-1}).

Used throughout the lectures (units in L03, L09, L10, L14-L16).

A number without a unit means nothing

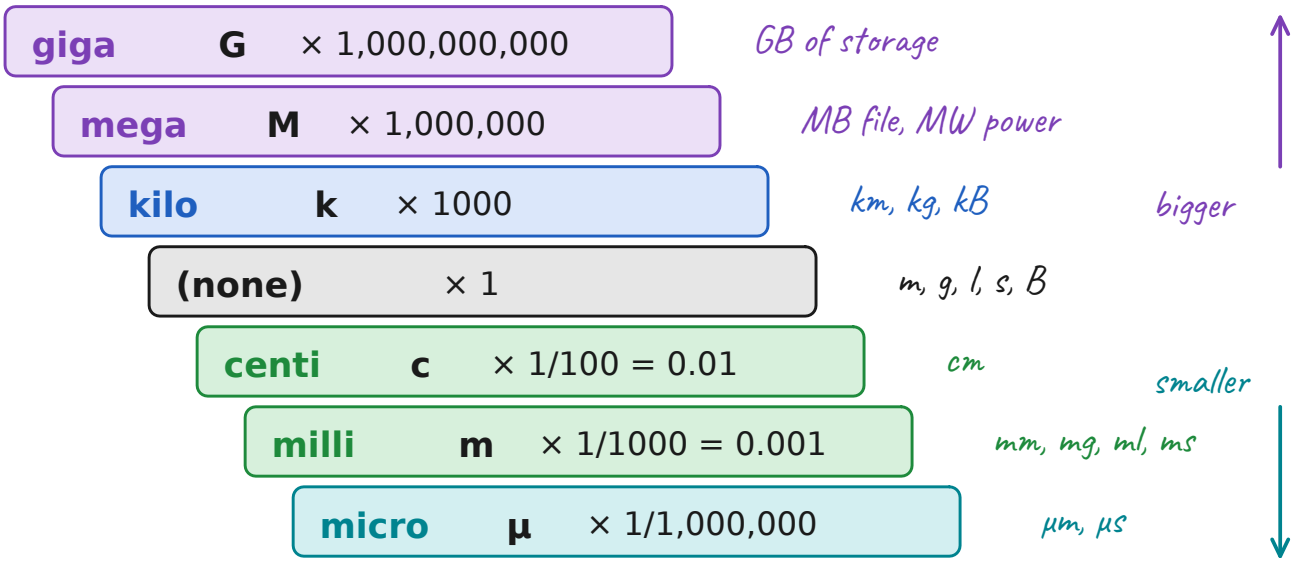
"The desk is 3 long." 3 what? Pencils? Hands? Metres? The **unit** is the thing you counted. **The number and the unit always travel together.**



Scientists agreed on standard units so everyone gets the same number: the **metre** (m) for length, the **kilogram** (kg) for mass, the **second** (s) for time, the **litre** (l) for liquid volume.

The prefix ladder: kilo, centi, milli...

Instead of inventing new names, the metric system sticks a **prefix** (a word-start) on the front. The prefix is a multiplier (Chapter 4). Each one means the same thing on **every** unit:



So a **kilo**gram is 1000 grams, a **milli**litre is a thousandth of a litre, a **centi**metre is a hundredth of a metre. **Careful:** capital M (mega) and small m (milli) differ by a factor of a thousand million!

Computer aside: in computing, a kilobyte is sometimes 1024 bytes instead of 1000, for reasons you'll see in Chapter 10 (number bases).

Changing units is like swapping coins

RULE: changing length, mass and capacity units

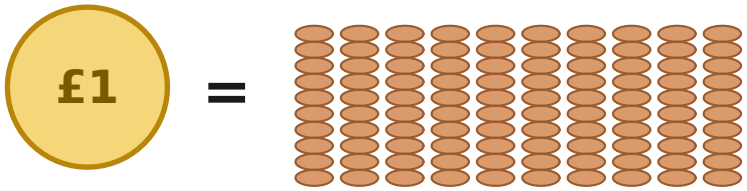
Know the swap rate: $1\ \text{km} = 1000\ \text{m}$, $1\ \text{m} = 100\ \text{cm}$, $1\ \text{cm} = 10\ \text{mm}$, $1\ \text{kg} = 1000\ \text{g}$, $1\ \text{l} = 1000\ \text{ml}$.

Big unit to small unit: multiply by the rate (the number gets bigger).

Small unit to big unit: divide by the rate (the number gets smaller).

Several steps: do one swap at a time along the chain.

A £1 coin is always worth exactly 100 pennies. Not roughly, not today's rate: **always**, by definition. Swap one pound for pennies and the **number gets bigger** (1 becomes 100), but you have exactly the same money, just counted in smaller pieces. Units work the same way: **1 m is always exactly 100 cm**.

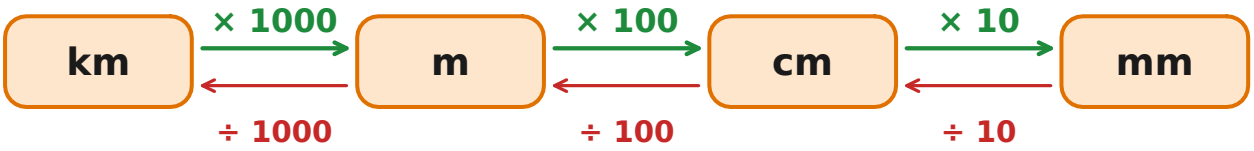


100 pennies (10 stacks of 10)

same amount of money, bigger number, because each piece is smaller. 1 m → 100 cm works identically.

(Currency exchange between countries is a bad model for this, because those rates change every day. Unit swaps never change: 1 km is 1000 m today, tomorrow and forever.)

going to smaller units: multiply (you need more of them)



going to bigger units: divide (you need fewer of them)

Example: 2.5 km in centimetres. Walk along the chain, one swap at a time:

$$2.5\ \text{km} \quad \longleftarrow \text{start}$$

$$2.5 \times 1000 = 2500\ \text{m} \quad \longleftarrow \text{swap km → m: × 1000}$$

$$2500 \times 100 = 250{,}000\ \text{cm} \quad \longleftarrow \text{swap m → cm: × 100}$$

Backwards: 350 g in kg. Grams are small, kilograms big, so divide: $350 \div 1000 = 0.35$ kg.

Sense check every time: is the new number bigger or smaller? Metres to centimetres must get bigger (a small unit needs lots). If it shrank, you went the wrong way.

Perimeter: walk round the fence

RULE: perimeter

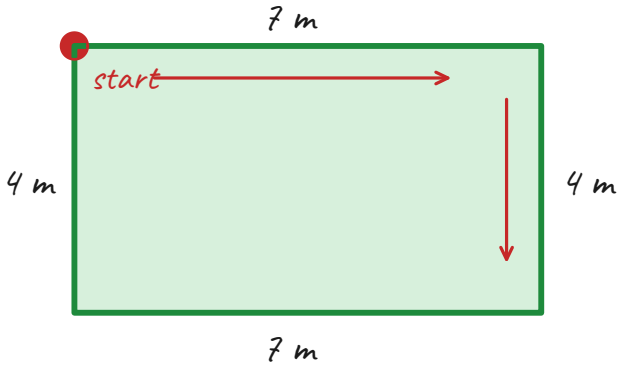
Perimeter = the sum of the lengths of **all** the sides. Its unit is a length (m, cm).

Rectangle, length l and width w: $P = 2l + 2w = 2(l + w)$. Square, side s: $P = 4s$.

Circle (circumference): $C = \pi d = 2\pi r$.

Shapes with missing sides: work the missing lengths out from the given ones first.

The **perimeter** is the distance all the way round the edge. Imagine walking along a garden fence and adding up every stretch until you're back at the start:

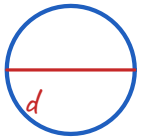


$$7 + 4 + 7 + 4 = 22 \text{ m}$$

every side counted once;
the answer is a length (m)

With the formula: $P = 2(l + w) = 2(7 + 4) = 2 \times 11 = 22$ m. Same answer, fewer additions.

Circles: the perimeter of a circle is called the **circumference**. Wrap a string round any circle and it's always about 3.14 diameters long; that number is π (Part 0):



$$C = \pi d = 2\pi r$$

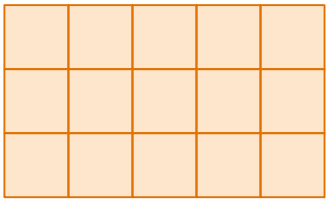
d = diameter (across), r = radius (centre to edge, half of d)

Area: count the squares

Area is how much flat surface a shape covers. It is measured in **square units** (cm², m²).

Shape	Area	Notes
Rectangle	$A = l \times w$	length × width
Square	$A = s^2$	side × side
Triangle	$A = \frac{1}{2}bh$	h is the **perpendicular** height, not the slant side
Parallelogram	$A = bh$	perpendicular height again
Trapezium	$A = \frac{1}{2}(a+b)h$	a, b = the two parallel sides
Circle	$A = \pi r^2$	r = radius (half the diameter)

A rectangle 5 cm by 3 cm holds 3 rows of 5 squares:

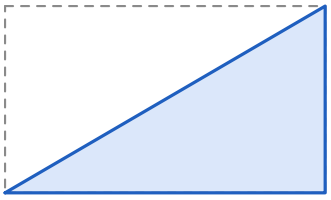


$$5 \times 3 = 15 \text{ cm}^2$$

cm² ("square centimetres"): each little square is 1 cm by 1 cm

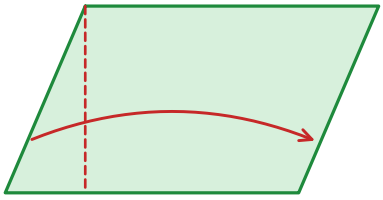
Other shapes are rectangles in disguise. **Cut and move pieces**; the area can't change:

triangle = half a rectangle



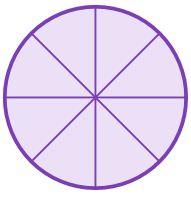
$$A = \frac{1}{2} \times b \times h$$

slide the corner across



$$A = b \times h$$

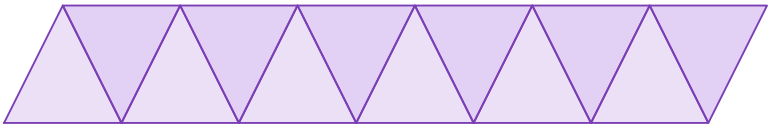
circle



$$A = \pi r^2$$

b = base, h = height measured straight up (at right angles to the base), not along the slope

Why πr² for a circle? Cut it into many thin pizza slices and lay them alternately point-up, point-down. They form an almost-rectangle: its height is the radius r and its length is half the circumference, πr. Area ≈ πr × r = πr².



height r, length πr

Trapezium and compound shapes

RULE: area of a trapezium

A trapezium has one pair of parallel sides, lengths a and b, a distance h apart.

$A = \frac{1}{2}(a+b)h$: average the parallel sides, multiply by the height.

Why: two copies, one turned upside down, make a parallelogram with base $a+b$ and height h.

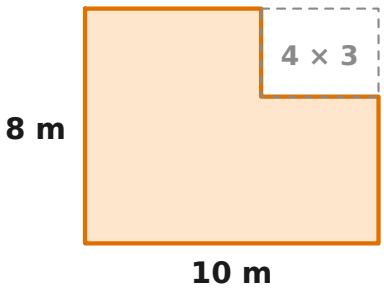
$$A = \tfrac{1}{2}(6+10) \times 4 = \tfrac{1}{2} \times 16 \times 4 = 32 \text{ cm}^2$$

parallel sides 6 cm and 10 cm, height 4 cm

PROCEDURE: compound shapes

1. Split the shape into rectangles, triangles, … (draw the cut lines).
2. Work out any missing side lengths from the ones you are given.
3. **Add** the areas of the pieces. For a hole or a corner cut out, **subtract** its area from the whole.
4. Perimeter: add only the **outside** edges (the cut lines are not part of it).

Example. The L-shape below: a 10 m by 8 m rectangle with a 4 m by 3 m corner removed.



whole rectangle $10 \times 8 = 80$ m^2

corner cut out $4 \times 3 = 12$ m^2

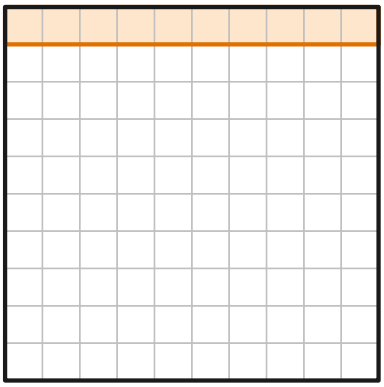
$$A = 80 - 12 = 68 \text{ m}^2$$

perimeter: still $2(10+8) = 36$ m (the step just moves two edges inward)

The big trap: 1 m² is NOT 100 cm²

It feels as if it should be. 1 m = 100 cm, so surely 1 m² = 100 cm²? Test it on a square small enough to see every piece: **one square centimetre**, blown up, with millimetres marked. 1 cm = 10 mm.

1 cm = 10 mm



1 cm
= 10 mm

orange: ONE row of 10 little squares. That's only 10 mm².

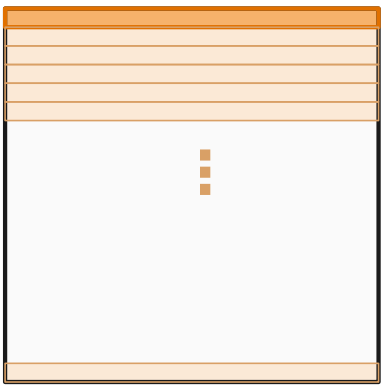
the whole square needs 10 rows of 10 = 100 little squares.

$$1 \text{ cm}^2 = 100 \text{ mm}^2$$

not 10 mm²: the 10 happens twice, once across and once down

Now the square metre. Same idea, bigger numbers. 100 cm² is **one ribbon** 100 cm long and 1 cm tall, running along the top edge. The square metre is 100 cm tall, so it needs **100 of those ribbons** stacked up:

1 m = 100 cm



1 m
= 100 cm
(100 ribbons)

top ribbon = 100 cm²
(100 little 1 cm squares in a row)

100 ribbons × 100 cm² each:

$$1 \text{ m}^2 = 10{,}000 \text{ cm}^2$$

ribbons drawn thicker than real, so you can see them

Why the little ² catches everyone. "m²" is short for m × m. So swap **each** m for 100 cm:

$$1 \text{ m}^2 = 1 \text{ m} \times 1 \text{ m} = 100 \text{ cm} \times 100 \text{ cm} = 10{,}000 \text{ cm}^2$$

The trap is mixing up two different things that look alike:

100 cm^2 ← *a hundred little 1 cm squares (one ribbon)*

$(100 \text{ cm})^2$ ← *a square whose SIDE is 100 cm: 10,000 little squares*

1 m² is the second one. The ² squares everything inside the bracket: the 100 **and** the cm.

RULE: converting area and volume units

If $1\ (\text{big unit}) = k\ (\text{small units})$ for length, then for **area** the rate is k^2 and for **volume** it is k^3.

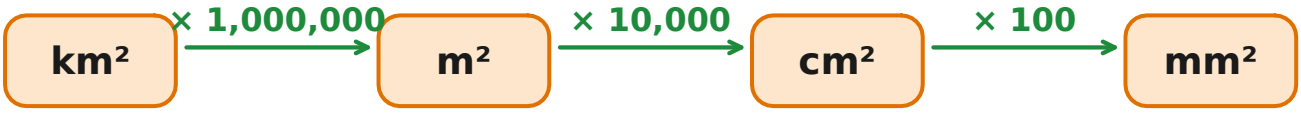
$1\ \text{m} = 100\ \text{cm}$, so $1\ \text{m}^2 = 100^2 = 10\,000\ \text{cm}^2$ and $1\ \text{m}^3 = 100^3 = 1\,000\,000\ \text{cm}^3$.

$1\ \text{cm} = 10\ \text{mm}$, so $1\ \text{cm}^2 = 100\ \text{mm}^2$ and $1\ \text{cm}^3 = 1000\ \text{mm}^3$ (so mm³ → cm³ is $\div 1000$).

Liquids: $1\ \text{l} = 1000\ \text{cm}^3$, $1\ \text{ml} = 1\ \text{cm}^3$, $1\ \text{m}^3 = 1000\ \text{l}$.

As before: big to small, multiply; small to big, divide.

The chain for areas:



Example: a room of 12 m² in cm²: $12 \times 10\,000 = 120\,000$ cm².

$$4500\ \text{mm}^3 = 4500 \div 1000 = 4.5\ \text{cm}^3$$ ← small to big: ÷ 10^3

$$0.25\ \text{m}^3 = 0.25 \times 1000 = 250\ \text{l}$$ ← $1\ \text{m}^3 = 1000$ litres

Volume: stack the cubes

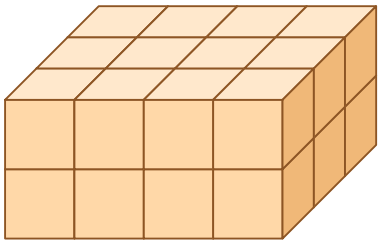
RULE: volume

Volume is measured in **cubic units** (cm³, m³).

Cuboid: $V = l \times w \times h$.

Any prism (same cross-section all the way through, e.g. a cylinder): $V = \text{area of cross-section} \times \text{length}$. Cylinder: $V = \pi r^2 h$.

Volume is how much 3-D space something fills, counted in unit cubes. A box 4 cm long, 3 cm wide and 2 cm tall: one layer on the floor holds 4 × 3 = 12 cubes, and there are 2 layers.



$$4 \times 3 \times 2 = 24 \text{ cm}^3$$

cm³ ("cubic centimetres"): count of 1 cm cubes

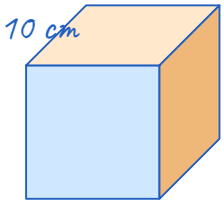
The layer idea works for any "straight-up" shape: volume = area of the floor × height. A cylinder (a tin can) is a stack of circles, like a pile of coins:

$$V = \pi r^2 \times h$$

floor area (a circle) × height.

r = 3 cm, h = 10 cm: V = π × 9 × 10 ≈ 282.7 cm³

Volume units, and litres. A 1 m cube holds 100 × 100 × 100 = 1,000,000 cm³ (the swap rate is cubed). Liquids use litres:



a 10 cm × 10 cm × 10 cm cube = 1000 cm³ = 1 litre

so 1 cm³ = 1 ml (a millilitre, a thousandth of a litre)

RULE: surface area of a cuboid

Surface area is the total area of all the faces (cm², m²): the cardboard needed to make the box.

A cuboid has 6 rectangular faces in 3 matching pairs:
$S = 2(lw + lh + wh)$.

A cube of side s: $S = 6s^2$ (and $V = s^3$).

Example. A box 8 cm by 5 cm by 2 cm.

$$V = 8 \times 5 \times 2 = 80 \text{ cm}^3$$

$$S = 2(8 \times 5 + 8 \times 2 + 5 \times 2) = 2(40 + 16 + 10) = 132 \text{ cm}^2$$

top/bottom, front/back, sides

Time is not decimal!

RULE: time

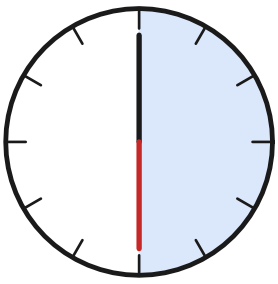
$1 \text{ h} = 60 \text{ min}$, $1 \text{ min} = 60 \text{ s}$, so $1 \text{ h} = 3600 \text{ s}$.

Decimal hours to h and min: keep the whole hours; minutes = decimal part $\times 60$. $2.75 \text{ h} = 2 \text{ h } 45 \text{ min}$.

h and min to decimal hours: hours + minutes $\div 60$.
$2 \text{ h } 15 \text{ min} = 2 + \frac{15}{60} = 2.25 \text{ h}$.

Adding times / durations: add minutes and hours separately; every 60 minutes carries 1 hour. Never use a calculator's decimal result as "hours.minutes".

An hour has **60** minutes, not 100. So 1.5 hours is **not** 1 hour 50 minutes. The ".5" means half an hour, which is 30 minutes. Look at the clock:



0.5 hour = half the clock = 30 min

0.5 h	0.5 × 60 = 30 min
0.25 h	0.25 × 60 = 15 min
0.1 h	0.1 × 60 = 6 min
2.75 h	2 h and 0.75 × 60 = 45 min

Hours → minutes: multiply the decimal part by 60. Minutes → hours: divide by 60, so 40 minutes is $\frac{40}{60} = \frac{2}{3} \approx 0.67$ hours.

Time arithmetic. A train leaves at 09:45 and the journey takes 2 h 35 min. When does it arrive?

$$9\text{ h }45\text{ min} + 2\text{ h }35\text{ min} = 11\text{ h }80\text{ min}$$ ← add hours, add minutes

$$80\text{ min} = 1\text{ h }20\text{ min} \Rightarrow 12{:}20$$ ← carry 60 min as 1 hour

Duration. From 10:50 to 13:25: count up to the next hour, then on.

$$10{:}50 \to 11{:}00 \to 13{:}00 \to 13{:}25: \quad 10 + 120 + 25 = 155\text{ min}$$

$$155\text{ min} = 2\text{ h }35\text{ min} = 2\tfrac{35}{60} \approx 2.58\text{ h}$$ ← 155 ÷ 60 = 2 r 35

✗ Common mistakes

✗ $2\text{ h }15\text{ min} = 2.15\text{ h}$ ✓ $2\text{ h }15\text{ min} = 2.25\text{ h}$

15 minutes is 15/60 = 0.25 of an hour, not 0.15

✗ $1.5\text{ h} = 1\text{ h }50\text{ min}$ ✓ $1.5\text{ h} = 1\text{ h }30\text{ min}$

0.5 h × 60 = 30 min

"Per" means divide: speed, density and friends

RULE: speed, distance, time (and density)

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \text{ so } \text{distance} = \text{speed} \times \text{time} \text{ and } \text{time} = \frac{\text{distance}}{\text{speed}}.$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}, \text{ so } \text{mass} = \text{density} \times \text{volume} \text{ and } \text{volume} = \frac{\text{mass}}{\text{density}}.$$

Units must match: km and hours give km/h; m and s give m/s. Convert minutes to hours (÷ 60) before using km/h.

A sprinter runs 100 m in 12.5 s. **Speed** is how many metres she covers **per** second: share the 100 m out equally among the 12.5 seconds.

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{100 \text{ m}}{12.5 \text{ s}} = 8 \text{ m/s}$$

read m/s as "metres per second": the units divide too!

Three ways to write the same unit. Your lecture slides write metres per second as "ms^{-1}" (Lecture 3 even writes "5ms-1"). It means the same as m/s, **but strictly it needs a space**: "m s^{-1}". Without the space, "ms" is the symbol for **milliseconds**, so "ms^{-1}" would mean "per millisecond". Write it with a space (or as m/s) in your own work:

$$\text{m/s} \qquad \frac{\text{m}}{\text{s}} \qquad \text{m s}^{-1}$$

the little $^{-1}$ means "per"; Chapter 9 explains why

Changing "per" units: one swap at a time. 8 m/s in km/h (kilometres per hour):

$$8 \text{ m per second}$$ ← *start*

$$8 \times 3600 = 28{,}800 \text{ m per hour}$$ ← *1 hour = 3600 seconds: 3600 times as far*

$$28{,}800 \div 1000 = 28.8 \text{ km per hour}$$ ← *metres → km: ÷ 1000*

Density is mass per volume. A 50 cm³ block weighing 135 g has density $\frac{135}{50} = 2.7$ g/cm³ (that's aluminium). The other two versions of each formula are in the rule box above; Chapter 14 shows how to rearrange any formula.

Working backwards: find the missing measurement

PROCEDURE: finding an unknown length, radius or time

1. Write the formula that links what you know to what you want.
2. Put in the known numbers.
3. **Undo** the operations around the unknown: $\times$ by $\div$, squaring by square-rooting, π by dividing by π.
4. Check the units, and that the size is sensible.

Know	Want	Use
area A of a square	side	$s = \sqrt{A}$
area A and width w of a rectangle	length	$l = A \div w$
area A, base b of a triangle	height	$h = 2A \div b$
circumference C	radius	$r = \dfrac{C}{2\pi}$
area A of a circle	radius	$r = \sqrt{\dfrac{A}{\pi}}$
volume V, length l, width w	height	$h = V \div (l \times w)$
distance d, speed s	time	$t = d \div s$

Examples.

$s^2 = 49 \text{ cm}^2 \Rightarrow s = \sqrt{49} = 7 \text{ cm}$ ← square: undo squaring

$l \times 4 = 36 \Rightarrow l = 36 \div 4 = 9 \text{ m}$ ← rectangle 36 m², width 4 m

$2\pi r = 31.4 \Rightarrow r = \dfrac{31.4}{2\pi} \approx 5.0 \text{ cm}$ ← circumference 31.4 cm

$\pi r^2 = 50 \Rightarrow r = \sqrt{\dfrac{50}{\pi}} \approx 3.99 \text{ cm}$ ← circle area 50 cm²

$$t = \frac{150\ \text{km}}{60\ \text{km/h}} = 2.5\ \text{h} = 2\ \text{h}\ 30\ \text{min}$$ ← *150 km at 60 km/h*

$$d = 8\ \text{m/s} \times 45\ \text{s} = 360\ \text{m}$$ ← *distance = speed × time*

✗ Common mistakes

✗ $\pi r^2 = 50 \Rightarrow r = \frac{50}{\pi}$

✓ $r = \sqrt{\frac{50}{\pi}}$

dividing by π gives r^2, not r: take the square root last

✗ $\text{time} = \text{speed} \div \text{distance}$

✓ $\text{time} = \text{distance} \div \text{speed}$

check the units: km ÷ (km/h) = h

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Changing units	big unit to small: multiply; small to big: divide. 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm, 1 kg = 1000 g, 1 l = 1000 ml
Perimeter	$P = 2(l + w)$ sum of all the outside edges; circle $C = \pi d = 2\pi r$
Rectangle, triangle	$A = lw, \qquad A = \frac{1}{2}bh$ h perpendicular to b
Parallelogram, trapezium	$A = bh, \qquad A = \frac{1}{2}(a + b)h$
Circle	$A = \pi r^2$
Compound shapes	split up and add; subtract holes and cut-out corners
Volume	$V = lwh, \qquad V = \pi r^2 h$ prism: cross-section area × length
Surface area (cuboid)	$S = 2(lw + lh + wh)$
Area and volume units	square the length rate for area, cube it for volume: $1\ \text{m}^2 = 10^4\ \text{cm}^2$, $1\ \text{m}^3 = 10^6\ \text{cm}^3$
Litres	$1\ \text{l} = 1000\ \text{cm}^3$, $1\ \text{ml} = 1\ \text{cm}^3$, $1\ \text{m}^3 = 1000\ \text{l}$
Time	decimal part × 60 = minutes; minutes ÷ 60 = decimal hours; 1 h = 3600 s
Speed	$s = \frac{d}{t}, \quad d = st, \quad t = \frac{d}{s}$
Density	$\rho = \frac{m}{V}, \quad m = \rho V, \quad V = \frac{m}{\rho}$ ρ (rho) = density
Working backwards	write the formula, substitute, undo the operations around the unknown

📏 Tape measure: your turn

Answers under each group in grey. Draw the chain or the picture first!

Changing units

1. 3.2 km in m
2. 450 mm in m
3. 2.5 kg in g
4. 750 ml in litres
5. 0.004 s in ms

1. 3200 m 2. 0.45 m 3. 2500 g 4. 0.75 l 5. 4 ms

Shapes

6. Perimeter and area of a 9 cm by 6 cm rectangle
7. Area of a triangle, base 10 cm, height 7 cm
8. Circumference and area of a circle, radius 5 cm (1 d.p.)

6. 30 cm, 54 cm² 7. 35 cm² 8. 31.4 cm, 78.5 cm²

Area and volume units

9. 3 m² in cm²
10. Volume of a 5 × 4 × 3 cm box
11. 2 litres in cm³
12. 0.5 m³ in cm³

9. 30,000 cm² 10. 60 cm³ 11. 2000 cm³ 12. 500,000 cm³

Time and per

13. 2.4 hours in hours and minutes
14. 150 km in 2.5 h: speed?
15. 20 m/s in km/h

13. 2 h 24 min 14. 60 km/h 15. 72 km/h

More shapes, units and time

16. Area of a trapezium: parallel sides 5 cm and 9 cm, height 4 cm
17. A 12 m by 9 m rectangle with a 5 m by 4 m corner cut out: area?
18. Surface area of a 6 cm × 3 cm × 2 cm box
19. 6500 mm^3 in cm^3
20. 3 h 40 min in decimal hours (2 d.p.)
21. A bus leaves at 08:35 and takes 1 h 50 min. Arrival time?

16. 28 cm^2 (½ × 14 × 4) 17. 88 m^2 (108 − 20) 18. 72 cm^2 (2(18 + 12 + 6)) 19. 6.5 cm^3 (÷ 1000) 20. 3.67 h (3 + 40/60) 21. 10:25 (9 h 85 min = 10 h 25 min)

Find the unknown

22. A square has area 64 cm^2. How long is a side?
23. A rectangle has area 54 m^2 and width 6 m. How long is it?
24. A circle has circumference 62.8 cm. Find the radius (1 d.p.)
25. A circle has area 154 cm^2. Find the radius (1 d.p.)
26. How long does 210 km take at 84 km/h?
27. A box has volume 120 cm^3, length 6 cm and width 4 cm. Find its height

22. 8 cm (√64) 23. 9 m (54 ÷ 6) 24. 10.0 cm (62.8 ÷ 2π) 25. 7.0 cm (√(154 ÷ π)) 26. 2.5 h = 2 h 30 min (210 ÷ 84) 27. 5 cm (120 ÷ (6 × 4))